

Problem Sheet 5

1. The spectral density function $f(\omega)$ of a stationary time series model having autocovariance function $\{\gamma_k\}$ ($k = 0, 1, 2, \dots$) may be written as:

$$f(\omega) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} \gamma_k e^{-i\omega k}, \quad \text{for } -\pi < \omega < \pi \quad (1)$$

Stating any general properties of the autocorrelation function that you assume, show that $f(\omega)$ may be written equivalently as:

$$f(\omega) = \frac{\gamma_0}{2\pi} \left[1 + 2 \sum_{k=1}^{\infty} \rho_k \cos(k\omega) \right] \quad (2)$$

where ρ_k is the autocorrelation function of the time series. Deduce that:

$$f(\omega) = f(-\omega), \quad \text{for } -\pi < \omega < \pi \quad (3)$$

2. (a) Consider the simple moving-average filter $L\{W_t\} = \sum_{j=-q}^q g_j W_{t-j}$ with weights $g_j = \frac{1}{2q+1}$, $-q \leq j \leq q$. If $W_t = a + \beta t$ where a and β are non-zero constants, show that $L\{W_t\} = W_t$.
 (b) Now let $L\{e_t\} = X_t$, where e_t a zero mean white noise process with variance σ^2 . By finding the mean and variance of X_t describe the behaviour of X_t when q is large.
3. Consider the ARMA(p, q) process

$$\phi(B)X_t = \theta(B)e_t \quad (4)$$

where $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ and $\theta(B) = 1 - \theta_1 B - \dots - \theta_q B^q$, $\theta_q \neq 0$, $\phi_p \neq 0$.

- (a) Derive the form of the spectral density function $S(f)$ of $\{X_t\}$.
 (b) Hence, or otherwise, identify the stationary and invertible process $\{X_t\}$ having the spectral density

$$S(f) = \frac{17 - 8 \cos(2\pi f)}{13 - 12 \cos(2\pi f)} \quad (5)$$

obtaining values for any autoregressive parameters $\{\phi_j\}$, any moving average parameters $\{\theta_j\}$, and σ^2 .

Hint: Start by considering the moving average part (numerator).

4. Consider the AR(1) time series model $(1 - 0.8B)X_t = W_t$, where W_t is a Gaussian white noise process. Generate four realizations of this time series, $x_1, \dots, x_n$ for $n = 128, 512, 1024$, and 2048.
- (a) Compute and plot the periodogram in each case. Include the spectral density for the AR(1) model in each plot.
 (b) Repeat the same for the MA(1) model $X_t = (1 - 0.8B)W_t$.
5. Calculate the spectral density for the following time series models.
- (a) X_t , where $(1 - \frac{4\sqrt{2}}{5}B + (\frac{4}{5})^2 B^2)X_t = (1 - (\frac{4}{5})^2 B^2)W_t$, and W_t is $WN(0, 1)$.
 (b) Y_t , where $(1 - \frac{5}{6}B)Y_t = X_t$, and X_t is as defined above.
 (c) Simulate a realisation of the process Y_t , for $n = 1024$. Plot its periodogram and compare it to the theoretical spectral density.

6. The data `sunspotz` in the `library(astsa)` contains the biyearly smoothed (12-month moving average) number of sunspots from June 1749 to December 1978 with $n = 459$ points that were taken twice per year.
 - (a) Perform a periodogram analysis and discuss the existence of predominant periods.
 - (b) Fit an autoregressive model (of sufficiently high order) using a model selection method of your choice, then obtain the estimated rational spectral density (you can use the function `arma.spec()` in `library(astsa)`). Compare with the results from the nonparametric spectral estimator in part (a).
7. The levels of salt concentration known to have occurred over rows, corresponding to the average temperature levels for the soil science data, are in `salt` and `saltemp` in `library(astsa)`. Plot the series and then identify the dominant frequencies by performing separate spectral analyses on the two series. Include confidence intervals for the dominant frequencies and interpret your findings.